Inference on a single mean, standard deviaton known

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Estimating an unknown mean μ, assuming σ is known

Suppose we have a population of values for some quantitative variable X, and we know $\sigma = \sigma_X$ but not μ. We may

- know that our population is normal.
- be able to take an iid sample of size $n \geq 30$.

In either case, the Central Limit Theorem and the general experience of statisticians tells us we will have a sampling distribution for $\bar{X}$ that is approximately normal:

$$\bar{X} \sim \text{Norm}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \qquad \Rightarrow \qquad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \text{Norm}(0, 1).$$

Note: There are two standard deviations involved, here. There is σ, the population standard deviation, and $\sigma/\sqrt{n}$, the standard deviation of the sampling distribution of $\bar{X}$. Convention is that we give that of the sampling distribution another name, calling it the **standard error**. In this case, we are dealing with the sampling distribution of $\bar{X}$, so we call this second standard deviation the **standard error of the mean**, or

$$\text{SE}_{\bar{X}} = \frac{\sigma}{\sqrt{n}}.$$

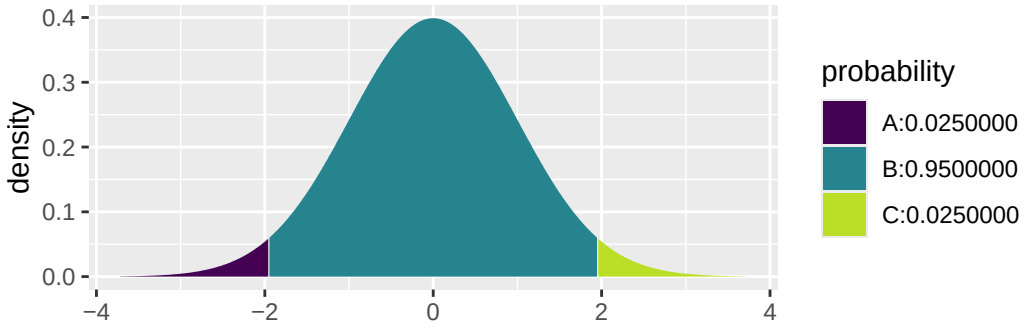
The command

```
xqnorm(c(0.025,0.975))
```

```
If X ~ N(0, 1), then

    P(X <= -1.959964) = 0.025   P(X <=  1.959964) = 0.975

    P(X >  -1.959964) = 0.975   P(X >   1.959964) = 0.025
```




```
[1] -1.959964  1.959964
```

displays that 95% of values in the standard normal distribution lie between $Z = -1.96$ and $Z = 1.96$. The Empirical Rule for normal distributions says, among other things, "approximately 95% of values lie within 2 standard deviations of the mean." Now we see the more accurate statement is that "95% of values lie within 1.96 standard deviations of the mean."

The natural implication is that 95% of values of $\bar{X}$ satisfy

$$-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96 \qquad \Rightarrow \qquad \bar{X} - (1.96)\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + (1.96)\frac{\sigma}{\sqrt{n}}.$$

We call the interval

$$(\bar{X} - (1.96)\frac{\sigma}{\sqrt{n}}, \bar{X} + (1.96)\frac{\sigma}{\sqrt{n}})$$

a 95% confidence interval for μ.

How to interpret this confidence

We are doing inference in constructing this interval. It involves drawing an iid sample, computing both the mean $\bar{X}$ and a **margin of error**

$$1.96\text{SE}_{\bar{X}} = 1.96\frac{\sigma}{\sqrt{n}}.$$

This margin of error is added and subtracted from a value, $\bar{X}$, that is sample-specific, in order to obtain lower and upper bounds on the interval. Thus, the generation of lower/upper bounds is the result of a random process. While μ is unknown, it is a fixed quantity which may or may not be contained between lower and upper bounds randomly generated.

In this context, what we are 95% confident about is the method of construction. We will gather just one sample (where randomness plays a role), and follow a deterministic method to obtain lower/upper bounds to an interval, and μ may or may not be inside that interval. We will not know for certain. We simply know this construction method is going to successfully "capture" μ in 95% of instances.

Other levels of confidence.

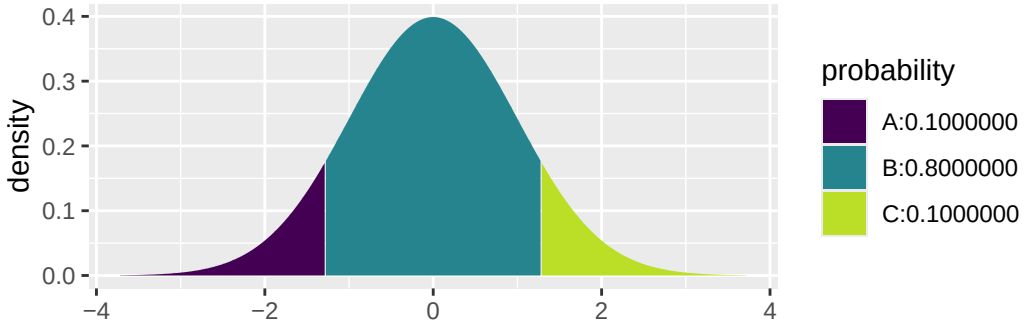
One can use `qnorm()` to capture different "interior areas". The nearest 80% of values (nearest to the mean) are represented as those between the 10th and 90th percentiles—those with Z-scores displayed here:

```
xqnorm(c(0.1,0.9))
```

```
If X ~ N(0, 1), then

    P(X <= -1.281552) = 0.1 P(X <=  1.281552) = 0.9

    P(X >  -1.281552) = 0.9 P(X >   1.281552) = 0.1
```




```
[1] -1.281552  1.281552
```

If we use $z^* = 1.2816$ in place of $z^* = 1.96$ in the calculation of the margin of error

$$1.2816\text{SE}_{\bar{X}} = 1.2816\frac{\sigma}{\sqrt{n}},$$

this change of method of construction changes the success rate to 80%. We can choose z^* to an desired level of success.

Hypothesis testing, σ known

In hypothesis testing, we ground things with the null hypothesis

$$\mathbf{H}_0 \colon \mu = \mu_0.$$

That is, we say something like, "μ is this value μ_0. Prove me wrong!" We may, actually, walk into the problem with prior experience that suggests $\mu > \mu_0$, and we wish to show this alternative is true:

$$\mathbf{H}_a \colon \mu > \mu_0.$$

We set the level of significance α, which dictates the probability

$$\alpha = \text{P}(\text{rejection of } \mathbf{H}_0 \mid \mathbf{H}_0 \text{ is true}),$$

and obtain evidence in the form of an iid random sample. The particular random sample generates an instance of a sample mean $\bar{X} = \bar{x}$. The p-value, as before, is

$$\mathrm{P}(\bar{X} > \bar{x} \mid \mathbf{H}_0 \text{ is true}) \;=\; \mathrm{P}\left(Z > \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right).$$

For 2-sided alternative hypotheses, this expression becomes$p$-value

$$\mathrm{P}\left(|Z| > \left|\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right|\right) \;=\; 2 \times \mathrm{P}\left(Z > \left|\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right|\right).$$

If $\alpha = 0.05$, the **rejection region** corresponds to $\bar{X}$-values

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < -1.96 \qquad \text{or} \qquad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} > 1.96.$$